



MASTER REPORT SCALING DataSAIL FOR LARGE DATASETS

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1. ABSTRACT

Machine learning in molecular property prediction has a unique challenge of models being highly sensitive to dataset splitting strategies. This is attributed to the presence of structural similarity between training and evaluation samples. Established splitting approaches such as random or scaffold-based splitting lead to information leakage, resulting in overly optimistic estimates of model generalization performance. DataSAIL was proposed as a leakage-aware splitting framework that minimizes this structural overlap between dataset partitions through similarity-based optimization. However, similarity computation and Integer Linear Programming (ILP) optimization impose computational complexity that limits its scalability for large molecular datasets.

We propose a scalable extension of DataSAIL based on representative subset optimization and similarity-based assignment. Instead of performing leakage-aware optimization on the complete dataset, the scaled approach uses a representative subset of molecules, selected using random or stratified sampling strategies. DataSAIL is then applied only to this subset to generate structurally coherent anchor partitions. The remaining molecules are subsequently assigned to train, validation, and test splits using a balanced k-nearest neighbour(kNN) similarity approach.

The balanced kNN assignment framework combines similarity-weighted split scoring along with balance-aware correction weights to preserve both structural consistency and target split ratios. The methodology subsequently evaluates trade-offs between computational efficiency, leakage reduction, runtime and split quality preservation.

To evaluate and assess the effectiveness of scaled DataSAIL, leakage analysis, split quality assessment, runtime evaluation, and downstream machine learning performance metrics such as Accuracy, F1 score, Matthews Correlation Coefficient (MCC), AUC, and PR-AUC, are used. Preliminary findings suggest that our scaled DataSAIL reduces computational cost while preserving much of the leakage-aware behaviour of full DataSAIL optimization. Initial findings are dataset dependent in terms of both runtime and leakage fraction.

2. INTRODUCTION

Machine Learning (ML) in molecular sciences has evolved from early cheminformatics in the 1990s and has become a dominant tool in molecular property prediction. It has expanded its reach to various applications, including but not limited to, molecular/drug discovery, toxicity screening, simulation, and chemical property estimation¹.

A critical aspect of this process is data splitting. Standard splitting strategies, such as random splitting, operate from the assumption that data points are independently and identically distributed. However, the assumption is often violated in molecular datasets. This is attributed to the fact that molecular datasets are often heavily imbalanced² and exhibit a high degree of autocorrelation³ (structural similarity). This results in very similar molecules existing in both train and test data, allowing models to rely on memorization instead of generalization, leading to high test accuracy that isn't truly reflective of actual predictive power for new, unseen data points.

This leads to data leakage or information leakage, a phenomenon where information from the training set effectively "leaks" into the test set through structural similarity.

Consequently, evaluation metrics can be overly optimistic and fail to reflect real-world performance on novel molecules⁴⁻⁶.

Data leakage, specifically, similarity-induced leakage, has been addressed by proposing similarity-aware splitting strategies⁶. Among these, the DataSAIL formulates dataset splitting as an optimization problem, aiming to minimize inter-split similarity while maintaining class balance and predefined split ratios. It reduces similarity-induced data leakage and helps the model generalize better to similarity aware out-of-distribution (OOD) settings.

DataSAIL provides a principled and effective solution. However, it encounters several computational challenges on large-scale datasets. Since it requires the computation of pairwise similarities between all molecules, the complexity sees a quadratic increase with respect to dataset size. Hence, the computational complexity becomes astronomical with increasing dataset size. In addition to that, the splitting procedure is formulated as an integer linear program (ILP). DataSAIL first clusters the data and then solves a constant-sized ILP heuristic over clusters. The DataSAIL optimization problem is NP-hard. This process becomes increasingly expensive as the dataset grows. Consequently, there is a need to scale DataSAIL to retain the key benefits of similarity-aware splitting without incurring the said computational costs.

This limitation, thus, highlights the research gap: how to retain the benefits of DataSAIL, that is, the reduced information leakage and balanced evaluation, while ensuring scalability to larger datasets. To overcome this, we propose a scalable approximation framework of DataSAIL. The idea is to apply an optimization-based splitting method to a carefully selected subset of the data and then extending this split to the full dataset using a similarity-based assignment strategy. This approach balances the strengths of DataSAIL in capturing structural relationships while also avoiding its computational bottlenecks.

The main contributions of this work are the following:

- i.) A scalable approximation framework for similarity-aware data splitting in molecular machine learning.
- ii.) A systematic analysis of sampling strategies for subset selection with a focus on their impact on representativeness and class balance.
- iii.) A balanced k-nearest neighbour assignment method that mitigates bias towards dominant splits while preserving structural consistency.
- iv.) An empirical evaluation of trade-offs between data leakage, runtime, scalability, and class-balance across multiple datasets.

3. BACKGROUND

3.1 Data Leakage

In previous, influential work on leakage, Kaufman et al.⁷, has defined data leakage, or information leakage, as the unintended transfer of information between the test set and training set. In subsequent works, data leakage has been investigated and mitigation of data leakage is constantly being comprehensively explored. Leakage has been further formalized as a violation of the independence between training and evaluation data, allowing information that should remain exclusive to the test set to influence the model either directly or indirectly. Consequently, models may exploit hidden correlations or shared information rather than learning patterns that generalize reliably to unseen data, leading to overly optimistic performance estimates.

In practice, it is often subtle and unintentional, making it difficult to detect and prevent. Recent work by Kapoor et al.⁸, highlights leakage as a widespread issue across machine-learning-based science, contributing to reproducibility failures and overly optimistic evaluation results across multiple scientific domains. Leakage can be attributed to a multitude of sources, including but not limited to, data collection procedures, preprocessing errors, external data contamination, feature engineering errors, improper dataset splitting, etc⁸.

Thus, we categorize data leakage broadly into two categories: data-level leakage and pipeline-level leakage. Data-level leakage arises from the way datasets are constructed or split, for example, when duplicate or highly similar samples appear across both train and test splits. In contrast, pipeline-level leakage occurs when information from the test set is inadvertently used during model training, feature selection, or hyperparameter tuning. Both forms are known to severely distort model evaluation, but data-level leakage is particularly difficult to detect and prevent in structured scientific datasets.

3.2 Data Leakage in Molecular ML

Data leakage is especially a prevalent issue in molecular machine learning due to the inherent structure of chemical and biological data. Molecular datasets often exhibit strong autocorrelation, where structurally similar compounds tend to have similar physicochemical or biological properties. Consequently, randomly splitting such datasets frequently leads to highly similar molecules being present in both training and test sets. This causes implicit information sharing across splits and thus the evaluation metrics do not accurately reflect real-world generalization.

Such overlap violates the assumption of independent and identically distributed (IID) data and allows models to rely on memorization of structural patterns rather than learning generalizable relationships³. Consequently, evaluation metrics obtained under such conditions tend to overestimate model performance, particularly when the model is applied to out-of-distribution (OOD) data. The limitations of such evaluation strategies are evidently exposed in drug discovery, where exploring beyond the boundaries of the training set to novel chemical space and the ability of the model to generalize is indispensable and foundational to the concept. The degradation of model performance under out-of-distribution (OOD) conditions in drug discovery settings has been demonstrated by Fooladi et al.⁹.

Importantly, this issue is not limited to small molecule datasets. Similar forms of leakage arise in other domains of molecular machine learning, such as protein–protein interaction prediction, drug–target interaction modelling, and sequence-based learning, where biological entities share evolutionary or structural similarities. In these settings, similarity between sequences or interaction partners can lead to implicit information sharing across splits, further amplifying the risk of misleading evaluation. For example, a model trained on a kinase with 90% sequence identity to a test-set kinase can easily memorize the binding site interactions, leading to inflated accuracy. Similarly, homologous protein sequences appearing across different splits can lead to sequence homology leakage and artificially inflated evaluation performance^{4,5}.

4. RELATED WORK

4.1 Data Splitting Strategies

The chosen splitting strategy directly affects the extent of information leakage and the authenticity of performance estimation. The most commonly used approach is random splitting, where, as the name suggests, samples are assigned to training, validation, and test sets uniformly at random. Random splitting is computationally simple and is widely used. Although, random splitting often produces overly optimistic evaluation results in molecular datasets^{6,9}.

Several structure-aware splitting strategies have been proposed to tackle similarity-induced leakage. One of the most widely accepted approaches is scaffold splitting. It is popularized in molecular machine learning benchmarks such as MoleculeNet¹¹. Scaffold splitting partitions molecules according to their core structural backbones, for example, Bemis–Murcko scaffolds¹², ensuring that compounds sharing the same core structural framework are assigned to the same split. Compared to random splitting, this approach produces more challenging and realistic evaluation settings by reducing direct scaffold overlap between training and test data.

Clustering and similarity-aware splitting approaches have also been explored and popularized to better capture structural relationships within molecular datasets. Recent work by Guo et al.³ proposed UMAP-based clustering splits to generate more rigorous evaluation scenarios for virtual screening tasks. Such methods better reflect distributional shifts encountered in real-world applications by grouping structurally related compounds prior to dataset partitioning.

Recent studies have shown that scaffold-based splitting can continue to overestimate model performance when structurally similar compounds possess different scaffolds¹⁰. For example, benzene- and pyridine-containing compounds may be classified as different scaffolds despite sharing highly similar structural and physicochemical characteristics. As a result, scaffold splitting may still permit substantial similarity leakage between training and test sets. Molecular similarity measures such as the Tanimoto coefficient capture relationships that are not fully represented by scaffold identity alone, allowing similarity leakage to persist across partitions. These limitations highlight the necessity of balancing computational simplicity and the ability of existing splitting strategies to effectively minimize information leakage.

4.2 Similarity-Aware and Optimization-Based Splitting

To further reduce similarity-induced leakage, several studies have explored similarity-aware and optimization-based dataset splitting strategies. Unlike conventional splitting methods, these approaches explicitly incorporate molecular similarity into the splitting process in order to minimize information sharing between training and evaluation data. Such methods aim to produce dataset splits that better reflect realistic out-of-distribution (OOD) generalization scenarios while preserving balanced and practically usable partitions.

Among these approaches, DataSAIL⁶ is a leading approach. Instead of relying on hardcoded chemical rules (like scaffold trees), it evaluates structural relationships globally using similarity matrices. DataSAIL not only minimizes similarity but also enforces several constraints to ensure that the partitions can be used for downstream machine learning tasks. These include maintaining predefined split ratios and preserving class balance across partitions.

After computing pairwise similarities, DataSAIL groups structurally related molecules into clusters using similarity-aware clustering. These clusters are then used for optimization. ILP is applied to the clusters to assign clusters to train, validation or test sets while also enforcing desired split proportions and stratification constraints. Cluster assignments are then propagated back to infer final dataset partitions. This clustering-based formulation is less complex as compared to direct optimization over all molecules and also preserves leakage-aware partitioning of splits⁶.

Similar optimization-driven strategies have also been explored in related areas of machine learning and graph partitioning, where partition quality is optimized under multiple competing constraints^{13,14}. Optimization-based splitting methods still face significant computational challenges.

Molecular datasets, in particular, require the computation of pairwise similarities between all samples, resulting in quadratic complexity with respect to dataset size. Other domains such as protein sequence analysis can exploit specialized clustering algorithms (e.g., CD-HIT, MMseqs2) that avoid exhaustive pairwise comparisons. Since the splitting procedure is formulated as an integer linear program (ILP), which is NP-hard, it becomes increasingly expensive to solve as dataset size grows. Therefore, the direct application of DataSAIL becomes impractical for large-scale molecular datasets, motivating the need for scalable approximation strategies that preserve its key advantages while reducing computational cost.

4.3 Approximation and Sampling Strategies

Approximation and sampling strategies are used to improve scalability on large datasets. Processing the entire dataset directly is computationally expensive, so, these approaches aim to construct representative subsets that preserve important statistical or structural characteristics of the original data while considerably reducing computational cost^{15,16}.

Sampling strategies can thus significantly influence the quality of downstream dataset partitions. Random sampling is computationally simple but fails to preserve class distributions or adequately represent rare regions of chemical space. Stratified sampling methods maintain class balance across sampled subsets, while diversity-based approaches aim to maximize structural representativeness¹⁷.

4.4 Nearest-Neighbour Based Assignment

Nearest-neighbour methods are widely used in machine learning for classification, clustering, label propagation, and similarity-based assignment tasks^{18,19}. In molecular machine learning, nearest-neighbour approaches are particularly attractive due to their ability to exploit structural similarity relationships between compounds. Similarity-based assignment methods can therefore provide a computationally efficient mechanism for extending dataset partitions from a representative subset to larger datasets.

Several approaches have explored distance-weighted and similarity-weighted nearest-neighbour strategies, where assignments are influenced more strongly by nearby samples in feature space²⁰. Such methods are commonly used in semi-supervised learning and label propagation tasks to transfer information between structurally related samples.

Nearest-neighbour assignment methods exhibit important setbacks in imbalanced settings. Standard k-nearest neighbour approaches display bias toward dominant classes or larger partitions, particularly when local neighbourhoods are unevenly distributed²¹. In the context of dataset splitting, this can lead to assignment collapse, where the majority of samples are preferentially assigned to already overrepresented splits i.e., the training set. These limitations motivate the need for balanced similarity-aware assignment strategies that simultaneously preserve structural consistency and maintain desired split ratios.

4.5 Research Gap

Random splitting remains computationally simple and widely adopted but frequently produces overly optimistic evaluation results due to substantial similarity overlap between training and test sets^{3,6,9}. Scaffold-based approaches partially circumvent this issue by separating molecules according to shared structural frameworks^{11,12}. Scaffold splitting may still permit considerable similarity leakage when structurally related compounds possess different scaffolds¹⁰.

More advanced similarity-aware and optimization-based methods, such as DataSAIL⁶, address these limitations more explicitly by directly minimizing inter-split similarity while maintaining class balance and predefined split ratios. These approaches provide more realistic evaluation settings for out-of-distribution (OOD) generalization. However, their reliance on pairwise similarity computations and computationally expensive optimization procedures significantly limits scalability to large molecular datasets.

There remains a lack of scalable frameworks that simultaneously:

- preserve the leakage-reduction benefits of similarity-aware splitting,
- maintain balanced and practically usable partitions,
- and remain computationally feasible for large-scale molecular datasets.

This gap motivates the development of scalable approximation strategies for similarity-aware dataset splitting, which is the focus of this work.

5. METHODOLOGY

5.1 Overview of the Proposed Framework

We propose a scalable approximation for similarity-aware dataset splitting in molecular machine learning. We aim to preserve the leakage-reduction benefits of DataSAIL while addressing the computational limitations associated with large-scale datasets. Our approach achieves that by applying similarity-aware optimization directly to a representative dataset. We first construct a representative subset of the data and apply DataSAIL to obtain a high-quality leakage-aware partition and then extend the resulting split to the remaining samples using a balanced similarity-based assignment strategy.

An overview of the proposed pipeline is illustrated in Figure 1. The framework consists of three main steps: downsampling the dataset, DataSAIL on the representative subset, and assignment of the remaining samples to the generated partitions.

First, a representative subset of the original dataset is selected using sampling-based strategies. The primary objective of this step is to reduce computational complexity while retaining the statistical and structural characteristics of the original dataset. Different sampling strategies, including random and stratified sampling, were explored.

In the second stage, DataSAIL is applied exclusively to the selected subset. Since the subset is considerably smaller than the original dataset, the computational cost associated with pairwise similarity computation and optimization is reduced. The resulting subset serves as a structurally separated anchor split that preserves the core leakage-aware properties of similarity-aware splitting.

The remaining samples are then assigned to the generated partitions using a balanced k-nearest neighbour (kNN) assignment strategy. Assignment decisions are based on structural similarity to samples within the representative subset while additionally incorporating balance-aware weighting to preserve predefined split ratios. This helps us scale similarity-aware splitting to larger datasets without requiring computationally expensive optimization over the complete dataset.

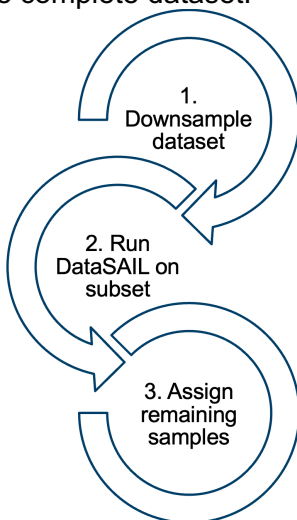


Figure 1- Hybrid Approximation approach for DataSAIL

The following sections describe each stage in detail.

5.2 Dataset Representation and Similarity Computation

In molecular ML, molecules are commonly represented using molecular fingerprints, which encode structural features into fixed-length binary vectors^{1,11}. We represented molecular structures using Morgan fingerprints²² generated through RDKit. Morgan fingerprints are circular fingerprints widely adopted in molecular property prediction and cheminformatics applications because they effectively capture local atomic environments and substructural patterns relevant for molecular similarity analysis.

We performed fingerprint generation using a radius of 2 and a fingerprint length of 2048 bits, corresponding to an Extended Connectivity Fingerprint (ECFP)-style representation^{22,23}. Molecules with invalid SMILES strings were excluded during preprocessing to ensure reliable fingerprint generation.

Structural similarity between molecules was assessed using binary fingerprint comparison. During nearest-neighbour assignment, similarity relationships were computed using the Jaccard distance²⁴ metric applied to Morgan fingerprints. For binary molecular fingerprints, Jaccard distance is mathematically equivalent to the

complement of the Tanimoto similarity coefficient^{23,24} and thus provides a computationally efficient approximation for similarity-based neighbour retrieval.

For two molecular fingerprints A and B , the Tanimoto similarity coefficient is defined as,

$$T(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

where $|A \cap B|$ denotes the number of shared fingerprint features and $|A \cup B|$ represents the total number of unique features across both fingerprints. Similarity values range from 0 to 1, where 1 indicates identical fingerprint representations and 0 indicates no shared structural features.

The Tanimoto coefficient has been extensively employed in virtual screening, similarity searching, and molecular retrieval tasks^{23,24}. Here, fingerprint similarity relationships were used during nearest-neighbour assignment of molecules outside the representative subset.

5.3 Subset Selection Strategy

To address the computational bottleneck associated with large-scale similarity-based dataset splitting, we introduce a subset selection strategy prior to the application of DataSAIL. Similarity-aware dataset splitting methods rely on pairwise similarity calculations between molecular samples. For a dataset containing n molecules, the similarity matrix requires,

$$O(n^2)$$

pairwise comparisons. As dataset size increases, both memory usage and runtime grow quadratically, making large-scale optimization computationally expensive.

To mitigate this scalability bottleneck, we adopt a representative approximation strategy. The assumption underlying this approach is that a sufficiently representative subset preserves the major structural and class-related properties of the full dataset. Thus, splits generated on the subset can approximate the global structure of the complete dataset while drastically reducing computational cost.

5.3.1 Sampling Methods

We employed two subset selection strategies in this work: random sampling and stratified sampling. Both approaches were investigated during methodological development and preliminary testing to produce a reduced subset. The representative subset size was 25% of the dataset.

Random Sampling

Random sampling selects molecules uniformly from the full dataset without considering class labels or structural properties.

Random sampling was implemented as a subset construction strategy. Given a dataset containing N molecules and sampling ratio r ,

$$n = r N$$

samples selected,

$$S \subseteq D$$

with,

$$|S| = n$$

It still preserved the original molecular indices for later reassignment. We implemented this approach by using NumPy uniform sampling without replacement. The procedure generates two partitions of a sampled subset used for DataSAIL optimization and a remaining subset to be assigned later through k-nearest neighbour propagation. This approach introduces minimal preprocessing overhead and is computationally inexpensive. This makes the approximation scalable and rapid.

Random sampling does not always work, particularly for imbalanced datasets like Tox21. Minority classes become underrepresented within the subset, which leads to biased split optimization and weaker preservation of class distributions during assignment. Since class labels are not considered during selection, the resulting subset does not preserve the original label distribution.

Stratified Sampling

We then implemented stratified sampling to determine whether preserving class distributions influenced subset quality. In this strategy, molecules are sampled proportionally according to their class labels, ensuring that the selected subset maintains approximately the same class distribution as the original dataset.

Stratified sampling was explored because leakage-aware splitting should preserve both structural separation and label balance, particularly in imbalanced datasets such as Tox21. Tox21 contains multiple toxicity endpoints; positive labels are the minority class. Maintaining representative class proportions within the subset improves the stability of downstream evaluation and reduces the likelihood of underrepresentation of minority classes.

For a dataset containing label groups $c_1, c_2, \dots, c_k$, molecules are sampled proportionally from each class according to the sampling ratio r . Thus, the resulting subset maintains the same class distribution of the original dataset. Stratified sampling was implemented using class-wise proportional sampling to preserve the prevalence of each class within the representative subset.

5.4 DataSAIL on Representative Subset

After constructing a representative subset, DataSAIL was applied exclusively to this reduced dataset in order to generate leakage-aware train, validation, and test splits. Rather than performing DataSAIL optimization over all molecules, only the selected subset underwent similarity-aware partition optimization. Since similarity matrix construction scales quadratically with dataset size²⁵, restricting optimization to a smaller subset provided substantial computational savings.

5.5 Extension of Splits to the Full Dataset

Once leakage-aware anchor partitions were generated on the representative subset, the next step is to assign the remaining molecules in the dataset to one of the existing splits. This stage is decisive because the quality of the final dataset partition depends not only on the subset optimization itself, but also on how accurately unseen molecules are propagated into the corresponding train, validation, or test partitions.

The assignment procedure was designed to preserve structural consistency with the subset splits while simultaneously maintaining balanced split sizes and class distributions. To achieve this, we explored the following assignment strategies.

5.5.1 Naïve Similarity-based Assignment

The first strategy for assigning unseen molecules we employed was naïve nearest-neighbour similarity propagation. For each molecule outside the representative subset, molecular similarity was computed against molecules contained within the anchor partitions. The molecule was then assigned to the split corresponding to its most similar neighbours. This approach is related to classical nearest-neighbour classification methods¹⁸, where labels are derived from neighbouring samples in feature space.

Employing a simple unweighted majority assignment strategy, the k nearest neighbours of a molecule were identified, and the molecule was assigned to the split receiving the highest number of neighbour votes. This ensures that structurally similar molecules are assigned to the same partition in order to preserve the structural organization established by the DataSAIL subset optimization. We implemented Naïve kNN using `sklearn.neighbors.NearestNeighbors` with `metric='jaccard'`.

A severe problem emerged during experimentation. Nearest-neighbour assignment showed a tendency toward train split dominance (Figure C.2). Since the training partition typically contains the largest number of molecules, unseen molecules are statistically more likely to encounter nearest neighbours originating from the train split. During iterative assignment, this imbalance gradually compounds, causing disproportionate growth of the training partition. Simple majority voting ignores global split balance constraints and validation and test partitions become severely underrepresented, leading to substantial deviations from the intended split ratios. This behaviour remains particularly problematic for minority classes and structurally sparse regions of chemical space. Molecules from these regions are frequently absorbed into the dominant split, reducing class diversity in validation and test partitions and weakening the reliability of downstream evaluation metrics²¹. These observations indicate that similarity preservation alone is insufficient and that explicit balance correction is required during assignment.

To address these issues, we developed a balance-aware assignment strategy designed to preserve structural similarity while maintaining desired split proportions.

5.5.2 Balanced k -Nearest Neighbour Assignment

To address the issues of naïve nearest-neighbour propagation, we introduced a balanced kNN assignment strategy. The method is designed to preserve structural consistency with the anchor partitions while maintaining target split proportions. Unlike naïve nearest-neighbour propagation, the balanced approach incorporates global

partition information during assignment, thereby limiting uncontrolled growth of dominant splits.

The proposed approach, shown in Figure 2, combines similarity-weighted split scoring with balance-aware weighting. Rather than assigning molecules solely based on neighbour frequency, neighbour contributions are weighted according to both molecular similarity and the current state of split balance.

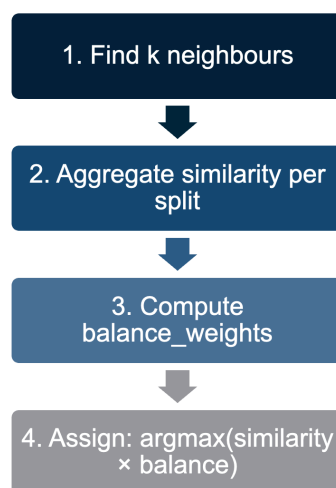


Figure 2- Balanced kNN Assignment

We implemented balanced kNN using the same neighbour-search process as naïve kNN and then implemented balanced scoring manually through custom implementation using NumPy and Pandas. Sklearn retrieves the neighbours and then neighbours are grouped by split. Mean similarity is then computed per split. Next, we apply balance correction and select the highest score.

This approach keeps in mind that similarity alone is insufficient because repeated assignment to already dominant splits gradually destroys the intended train-validation-test proportions. To mitigate this issue, we adjusted similarity-based assignment using balance-aware correction factors. Splits that were underrepresented relative to their target ratios received higher weighting, whereas oversized splits were penalized. This enabled assignment decisions to adapt dynamically throughout the propagation process.

Similarity-Based Split Scoring

For each of the unassigned molecules, we identified the k nearest neighbours among the subset molecules using fingerprint similarity measures^{22–24}. The retrieved neighbours were grouped according to their assigned partition (train, validation, or test).

We computed the similarity score using `sklearn.neighbours.NearestNeighbours` with Jaccard distance using,

$$\text{Similarity} = 1 - \text{Jaccard Distance}$$

For each candidate split, the similarity contribution was computed as the mean similarity of neighbouring molecules belonging to that split. Splits containing neighbours that were more structurally similar to the query molecule received higher

similarity scores. The resulting split-specific similarity scores were then combined with balance-aware correction factor to determine the final assignment decision.

Balance-Aware Weighting

To preserve the intended dataset proportions, we introduced a balance correction factor for each split. The balance factor was defined as:

$$BalanceFactor(s) = \frac{TargetRatio(s)}{CurrentRatio(s) + \epsilon}$$

where:

- $TargetRatio(s)$ denotes the desired proportion for split s ,
- $CurrentRatio(s)$ represents the current proportion after assignment,
- ϵ is a small stabilization constant preventing division instability.

The balance factor penalizes oversized splits while favouring underrepresented ones. And thus, assignment decisions dynamically adapt to maintain the global split distribution.

The final score for each candidate split was computed as:

$$Score(s) = SimilarityScore(s) \times BalanceFactor(s)^\lambda$$

Where λ denotes the balance-weight parameter controlling the influence of balance correction. The molecule was then assigned to the split achieving the highest final score.

The balance-weight parameter λ controls the trade-off between structural consistency and split balance. When $\lambda = 0$, assignment depends solely on molecular similarity and reduces to similarity-based propagation. Increasing λ strengthens the influence of balance correction, favouring underrepresented partitions. Excessively large λ values may increasingly prioritise balance over similarity information and potentially compromise structural consistency.

$\lambda = 2$ was used for all experiments. Sensitivity analyses conducted on the Tox21 and HIV datasets demonstrated that increasing λ improved adherence to the target split proportions while only modestly affecting leakage metrics. Consequently, $\lambda = 2$ was selected as a practical compromise between preserving structural separation and maintaining balanced split sizes.

Algorithm Description

We summarize the balanced assignment procedure as follows:

1. Compute molecular fingerprints for all molecules.
2. Generate anchor partitions on the representative subset using DataSAIL.
3. For each remaining molecule:
 - Identify the $k = 5$ nearest subset neighbours,
 - Compute similarity-weighted split scores,
 - Apply balance-aware correction factors,
 - Assign the molecule to the highest-scoring split.
4. Update split ratios dynamically after each assignment.

This iterative propagation procedure helps us preserve both local structural organization and global split balance simultaneously.

Computational Complexity

The proposed framework substantially improves scalability compared to full-dataset optimization. For a dataset containing n molecules and a representative subset of size m , where $m \ll n$, DataSAIL optimization is performed only on the subset. Similarity computation and ILP optimization scale with:

$$O(m^2)$$

instead of:

$$O(n^2)$$

After subset optimization, the remaining molecules are assigned using balanced k-Nearest Neighbour propagation. For k neighbours and $n - m$ unassigned molecules, assignment scales as,

$$O((N - m) \cdot k)$$

The total asymptotic complexity of the approach is,

$$O(m^2 + (N - m) \cdot k)$$

The exact complexity of neighbour retrieval depends on the nearest-neighbour implementation. In our implementation, neighbour search was performed using *sklearn.neighbors.NearestNeighbors* with *Jaccard* distance applied to *Morgan fingerprints*. The assignment stage scales substantially more favourably than full pairwise optimization and approximately with the number of remaining molecules and neighbour comparisons, making it more computationally feasible for large datasets.

5.6 Evaluation Framework

To assess the effectiveness of the proposed scalable splitting methodology, we considered multiple evaluation criteria. Evaluation focused not only on downstream machine learning performance, but also on leakage reduction, split quality preservation, and computational scalability.

5.6.1 Leakage Metrics

A primary objective of leakage-aware splitting is to minimize structural overlap between train, validation, and test partitions⁶⁻⁸. We did this using following metrics.

Mean Similarity- Mean inter-split similarity measures the average structural similarity between molecules belonging to different partitions. This metric corresponds to the inter-split similarity formulation used in DataSAIL (Equation 2)⁶ which aggregates similarity between molecules assigned to different splits. Mean similarity was computed using pairwise fingerprint comparisons between molecules in different splits. Molecular similarity was evaluated using Morgan fingerprints and Tanimoto-equivalent similarity computation based on Jaccard comparison²²⁻²⁴. Lower mean similarity

indicates improved separation between train and test chemical spaces, thereby reducing the likelihood of information leakage.

$$MeanSimilarity = \frac{1}{|D_{train}| \times |D_{test}|} \times \sum_{x \in D_{train}} \sum_{x' \in D_{test}} sim(x, x')$$

Nearest-Neighbour Leakage- Nearest-neighbour leakage evaluates whether highly similar molecules appear across different partitions. This metric follows the nearest-neighbour leakage formulation proposed by Elangovan et al.²⁵ and adopted within DataSAIL (Equation 1). For each molecule in the test set, the similarity to its nearest neighbour is computed. High nearest-neighbour similarity values indicate potential leakage, as structurally related compounds allow predictive models to rely on memorized chemical patterns rather than learning features that generalize to unseen molecules¹⁰. Nearest Neighbour was implemented using sklearn’s *NearestNeighbours* with Jaccard distance applied to Morgan fingerprints.

$$NNLeakage = \frac{1}{|D_{test}|} \times \sum_{x \in D_{test}} \max_{x' \in D_{train}} sim(x, x')$$

Leakage Fraction- Leakage fraction measures the proportion of test molecules whose most similar training molecule exceeds a chosen similarity threshold. Higher leakage fractions therefore indicate greater structural overlap between splits and provide a convenient basis for comparing alternative splitting strategies. To implement leakage fraction, we used nearest-neighbour similarity relationships along with threshold-based filtering. Molecules exceeding the similarity threshold of $\tau = 0.7$ (exploratory threshold choice) across train-test boundaries were classified as leaked samples.

$$LF = \frac{1}{|D_{test}|} \times \sum_{x \in D_{test}} \mathbb{1} \left(\max_{x' \in D_{train}} sim(x, x') > \tau \right)$$

5.6.2 Split Quality Metrics

Reducing similarity-induced leakage alone is insufficient for evaluating the quality of generated dataset splits. High-quality splits should preserve both the statistical characteristics of the original dataset and the intended partition structure, while maintaining the structural properties of the underlying chemical space. We, thus, evaluated the split using three metrics: statistical representativeness, ratio preservation, and structural representativeness.

Statistical Representativeness- Statistical representativeness assesses if generated splits preserve the label distribution of the original dataset faithfully. This aspect is particularly important for imbalanced molecular datasets such as Tox21, where distorted class proportions within the train, validation, or test partitions produce significant bias during downstream evaluation and affect the interpretation of predictive performance^{2,21}.

We assessed class distribution preservation using the **Distribution Gap (DG)**, defined as the absolute difference between the prevalence of the positive class in a given split and the prevalence observed in the complete dataset:

$$DG_i = |p_i - p|$$

where p denotes the positive class prevalence in the original dataset and p_i denotes the prevalence within split i .

Because absolute deviations may underestimate the impact of distribution shifts in highly imbalanced datasets, we computed the **Distribution Ratio Error (DRE)**:

$$DRE_i = \frac{|p_i - p|}{p}$$

where lower values indicate better preservation of the original class prevalence.

We further quantified statistical fidelity using the **Kullback-Leibler (KL) divergence** between the original label distribution and the split-specific distribution. For binary classification tasks, the divergence was computed as:

$$D_{KL}(P \parallel Q_i) = p \ln\left(\frac{p}{q_i}\right) + (1 - p) \ln\left(\frac{1 - p}{1 - q_i}\right)$$

Where,

$$P = (p, 1 - p)$$

represents the original dataset distribution and,

$$Q_i = (q_i, 1 - q_i)$$

represents the label distribution within split i . Lower KL divergence values indicate greater agreement between the generated splits and the original dataset.

Ratio Preservation- Ratio preservation evaluates how accurately the intended train-validation-test proportions are maintained after assignment. Since similarity propagation gradually shifts the partition sizes away from the target split ratios, this metric assesses the effectiveness of the balance-aware assignment strategy. Deviations from the desired 70:20:10 proportions were quantified by comparing the observed partition sizes with the predefined targets. Lower deviations therefore indicate improved adherence to the intended split structure.

Structural Representativeness- Preservation of label distributions alone does not guarantee that representative subsets adequately capture the structural characteristics of the original molecular dataset. We therefore assessed the structural representativeness by comparing the mean pairwise molecular similarities of the full datasets and their corresponding representative subsets.

For a dataset containing (n) molecules, the mean pairwise similarity is defined as,

$$S(D) = \frac{2}{n(n - 1)} \sum_{\{i < j\}} sim(x_i, x_j)$$

where $sim(x_i, x_j)$ denotes the Tanimoto similarity between the fingerprints of molecules x_i , and x_j . For a representative subset containing m molecules, the corresponding mean similarity is computed as,

$$S(D) = \frac{2}{m(m-1)} \sum_{\{i < j\}} \text{sim}(x_i, x_j)$$

Structural preservation is then quantified as,

$$\Delta S = |S(D) - S(S)|$$

Lower values of ΔS indicate that the representative subset preserves the structural characteristics of the original chemical space. Exact pairwise similarities were computed for smaller datasets, whereas for HIV, mean similarities were estimated using 100,000 randomly sampled molecular pairs to maintain computational feasibility.

5.6.3 Machine Learning Evaluation

The practical utility of our approach is assessed through downstream molecular property prediction experiments using standard machine learning evaluation metrics¹¹. These methods were implemented using sklearn evaluation utilities.

Accuracy- Accuracy measures the proportion of correctly classified samples among all predictions. Although widely used, accuracy alone might be misleading for imbalanced datasets because dominant classes disproportionately influence the metric.

F1 Score- The F1 score represents the harmonic mean of precision and recall, balancing false positives and false negatives. This metric is useful when class distributions are uneven.

Matthews Correlation Coefficient (MCC)- Matthews Correlation Coefficient provides a balanced evaluation measure even under severe class imbalance². Unlike accuracy, MCC encompasses all elements of the confusion matrix and therefore provides a more reliable assessment of predictive quality for molecular datasets with skewed label distributions.

Area Under the ROC Curve (AUC)- AUC assesses the ability of a classifier to distinguish between positive and negative classes across different decision thresholds. Higher ROC-AUC values indicate stronger separability between positive and negative samples.

Precision-Recall Area Under Curve (PR-AUC)- PR-AUC is informative for highly imbalanced datasets because it focuses on precision and recall rather than true negative rates. For molecular property prediction tasks involving rare positive samples, PR-AUC often provides a more realistic assessment of model utility.

5.6.4 Runtime Evaluation

Runtime analysis compares the computational cost of full DataSAIL optimization against the proposed subset-based framework.

The evaluation focused on:

- similarity computation time,
- subset optimization runtime,

- assignment runtime,
- total pipeline execution time.

This analysis enabled investigation of the trade-off between computational efficiency and leakage reduction performance.

6. RESULTS

The experiments aimed to assess whether the proposed scalable approximation preserved the desirable properties of DataSAIL while improving computational feasibility. We evaluated the extent to which leakage reduction benefits of DataSAIL were preserved, whether split quality characteristics were preserved, the impact on downstream machine learning performance, and the computational efficiency, specifically, runtime of our approach. Where applicable, results obtained using random and stratified downsampling were compared. The reported results correspond to a single experimental run using a fixed random seed (seed = 42) and should therefore be interpreted as exploratory observations.

6.1 Leakage Analysis

To evaluate leakage-reduction, we considered three metrics, i.e., mean train-test similarity, nearest-neighbour leakage, and leakage fraction. Lower values for all three metrics indicate improved structural separation between training and test partitions and to that extent reduced potential for similarity-induced information leakage.

Mean Train-Test Similarity

The results are summarized in Table 1. For the BACE dataset, random splitting produced the highest mean train-test similarity (0.167), indicating substantial structural overlap between the partitions. Full DataSAIL reduced this value to 0.123, while DataSAIL scaled achieved even lower values (0.107 for random downsampling and 0.108 for stratified downsampling). Thus, the scalable approximation slightly exceeded the structural separation achieved by full optimization.

For BBBP, differences between splitting strategies were comparatively small. Mean train-test similarities ranged between 0.100 and 0.104, suggesting that the dataset exhibits relatively limited sensitivity to the choice of leakage-aware splitting strategy. Stratified scaled DataSAIL produced the lowest value (0.100), although improvements over random splitting were modest. For Tox21, random splitting yielded a mean similarity of 0.084, whereas both full DataSAIL and scaled DataSAIL reduced this value to 0.064.

Thus, mean train-test similarity analyses suggest that our approach largely preserved the structural separation properties of DataSAIL while extending evaluation to datasets for which full optimization was impractical.

dataset	split	mean_train_test_similarity	train_sample_size	test_sample_size
bace	random	0.167	1059	152
bace	datasail_full	0.123	1040	150
bace	datasail_scaled_random	0.107	1025	170
bace	datasail_scaled_stratified	0.108	1042	163
bbbp	random	0.104	1426	205
bbbp	datasail_full	0.101	1460	203
bbbp	datasail_scaled_random	0.103	1423	207
bbbp	datasail_scaled_stratified	0.1	1411	202
tox21	random	0.084	5475	783
tox21	datasail_full	0.064	5684	704
tox21	datasail_scaled_random	0.064	5438	798
tox21	datasail_scaled_stratified	0.061	5467	786

Table 1- Mean train-test similarity for BACE, BBBP, and Tox21

Nearest-Neighbour Leakage

Table 2 shows that across all datasets, random splitting consistently produced the highest nearest-neighbour similarities. For BACE, the mean maximum similarity decreased substantially from 0.789 under random splitting to 0.344 using full DataSAIL. The scaled approach closely reproduced this behaviour, yielding values of 0.362 and 0.367 for random and stratified downsampling, respectively.

BBBP exhibited a similar trend, although the improvements were less pronounced. Random splitting resulted in a mean nearest-neighbour similarity of 0.560, whereas full DataSAIL reduced this value to 0.313. The scaled variant achieved intermediate values of 0.437 and 0.425, suggesting partial preservation of DataSAIL's leakage-reduction behaviour.

For Tox21, scaled approach observed the greatest reduction. Mean maximum similarity declined from 0.570 under random splitting to 0.411 in full DataSAIL and 0.394 in the scaled DataSAIL. Amongst the two scaled approaches, stratified downsampling proved particularly effective with mean maximum similarity of 0.394 as compared to the 0.486 of random downsampling approach.

Leakage Fraction

Random splitting consistently exhibited the highest leakage fractions. The strongest benefit of stratification was observed for Tox21. Leakage fraction declined from 24.1% with random splitting to 1.1% using full DataSAIL. Stratified scaled DataSAIL retained much of this behaviour, yielding a leakage fraction of 4.2%.

In BACE, 84.9% of test molecules exceeded the leakage threshold. Full DataSAIL reduced this proportion dramatically to 7.3%, while the scaled approaches achieved even lower fractions of 4.1% and 1.2% for random and stratified downsampling, respectively. In BBBP, leakage fraction decreased from 27.8% under random splitting to 2.0% using full DataSAIL. The scalable approximations produced intermediate values of 9.2% and 9.4%, substantially improving upon the random baseline while not fully matching the leakage reduction achieved by complete optimization.

This shows that our scaled framework substantially reduced the proportion of potentially leaked test samples relative to conventional random splitting. While full DataSAIL generally achieved the strongest leakage reduction where computationally feasible, the scaled approximations preserved much of this behaviour. The relative performance of random and stratified downsampling depended on dataset characteristics, with stratification appearing particularly advantageous for the imbalanced Tox21 dataset but not universally superior.

dataset	split	mean_max_similarity	std_max_similarity	leakage_fraction
bace	random	0.789	0.105	0.849
bace	datasail_full	0.344	0.149	0.073
bace	datasail_scaled_random	0.362	0.121	0.041
bace	datasail_scaled_stratified	0.367	0.112	0.012
bbbp	random	0.56	0.213	0.278
bbbp	datasail_full	0.313	0.113	0.02
bbbp	datasail_scaled_random	0.437	0.169	0.092
bbbp	datasail_scaled_stratified	0.425	0.187	0.094
tox21	random	0.57	0.19	0.241
tox21	datasail_full	0.411	0.127	0.011
tox21	datasail_scaled_random	0.486	0.171	0.127
tox21	datasail_scaled_stratified	0.394	0.148	0.042

Table 2- NN Leakage and leakage fraction for BACE, BBBP and Tox21

6.2 Split Quality

We next observed whether the resulting splits preserved the statistical and structural characteristics of the original datasets. We evaluated the split quality in three ways: statistical representativeness, ratio preservation, and structural representativeness.

Statistical Representativeness

Final Split Distribution Preservation

We evaluated statistical representativeness of the final train, validation, and test partitions generated by Random splitting, full DataSAIL, and Scaled DataSAIL. Figure 3 shows that random splitting generally maintained the smallest train-test distribution differences because class proportions are preserved naturally through random allocation. However, both DataSAIL and Scaled DataSAIL introduced larger deviations as a consequence of prioritising structural separation and leakage reduction.

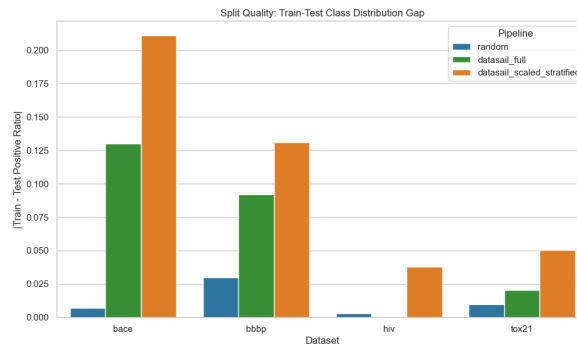


Figure 3- Split Quality Balance

Representative Subset Preservation

We additionally evaluated statistical representativeness to quantify the extent to which the class distributions within the generated partitions deviate from those observed in the complete dataset. This was done using distribution gap, distribution ratio error (DRE), and Kullback-Leibler (KL) divergence.

Distribution gap analyses in Table A.1 (Appendix A) reveal that the effect of stratification was highly dataset dependent. For Tox21 (using the NR-AR endpoint as a representative binary task), which exhibits substantial class imbalance, stratified

downsampling substantially improved preservation of label distributions. The mean distribution gap decreased from 0.043 under random downsampling to 0.018 under stratified downsampling, while the maximum deviation was reduced from 0.087 to 0.025. Similar, although more modest, improvements were observed for HIV, where the mean gap decreased from 0.017 to 0.016.

It was observed that stratification did not consistently improve distribution preservation for comparatively balanced datasets. For BBBP, the mean distribution gap increased from 0.045 to 0.067, while BACE exhibited nearly identical behaviour under both sampling strategies, with mean gaps of approximately 0.132. The large gaps observed for BACE reflect the sensitivity of small datasets to modest changes in class counts within the generated partitions.

Distribution Ratio Error (Table A.2) reiterates the importance of considering the underlying class imbalance. Since DRE normalises deviations relative to the prevalence of the positive class, improvements became particularly apparent for Tox21. Mean DRE decreased substantially from 1.024 under scaled random downsampling to 0.428 with stratified downsampling. HIV exhibited a similar trend, with mean DRE values decreasing from 0.491 to 0.443. Similar to the distribution gap, DRE for BBBP and BACE showed only marginal differences between the two sampling approaches, with random downsampling producing slightly lower DRE values.

KL divergence further supported these findings. Stratified downsampling yielded lower mean KL divergence (Table A.3) for HIV (0.0036 versus 0.0061) and Tox21 (0.0069 versus 0.0207), indicating improved agreement between the split-specific and original label distributions. For BACE and BBBP, however, random downsampling produced slightly smaller divergence values. Hence, the principal benefit of stratification lies in preserving class proportions within imbalanced datasets.

Ratio Preservation

Our balanced k-nearest neighbour assignment was designed to maintain the intended train-validation-test proportions during propagation of the remaining molecules. The resulting partition sizes remained close to the predefined target ratios across all datasets, indicating that the balance-aware weighting mechanism effectively mitigated the split dominance observed during preliminary experiments using naïve nearest-neighbour assignment.

Dataset	Train	Validation	Test	Max deviation
BACE	68.87%	20.36%	10.77%	±1.13 pp
BBBP	69.20%	20.89%	9.91%	±0.89 pp
Tox21	69.88%	20.07%	10.05%	±0.12 pp
HIV	70.24%	5.16%	10.08%	±0.32 pp

Table 3— Ratio preservation for DataSAIL scaled approach across 4 datasets

Structural Representativeness

We further assessed structural representativeness by comparing the mean pairwise similarities of the full datasets and their corresponding representative subsets.

Across all datasets, the representative subsets closely preserved the structural characteristics of the original molecular collections. For BACE, the absolute difference between the full dataset similarity and the subset similarity was 0.001 for random

sampling and 0.003 for stratified sampling. BBBP exhibited virtually no deviation between the original dataset and either subset construction strategy.

Similarly, Tox21 demonstrated only negligible structural differences, with delta similarities of 0.000 and 0.002 for random and stratified sampling, respectively. For HIV, despite its substantially larger size, both subset construction methods reproduced the mean pairwise similarity of the original dataset almost exactly, with no measurable deviation under random sampling and a difference of only 0.001 under stratification (Table A.4).

Therefore, the representative subsets comprising 25% of the original datasets were sufficient to preserve the broader structural characteristics of the molecular chemical space.

6.3 Machine Learning Performance

Next, to evaluate whether reduced structural leakage translated into changes in model behaviour, Random Forest and Logistic Regression classifiers were trained using the generated dataset partitions. Performance was assessed using Accuracy, F1 score, Matthews Correlation Coefficient (MCC), ROC-AUC, Precision, Recall, and PR-AUC. Here, we use MCC as primary metric for interpretation as it provides a balanced evaluation under class imbalance and incorporates all components of the confusion matrix.

Matthews Correlation Coefficient

Random splitting consistently produced the highest MCC values across all datasets. For BACE, the average MCC decreased from 0.585 under random splitting to 0.185 using full DataSAIL and 0.125 using scaled stratified DataSAIL. Similarly, BBBP exhibited a reduction from 0.620 with random splitting to 0.185 under full DataSAIL. However, the scaled approach retained considerably higher predictive performance of 0.458. Figure 4 illustrates the distribution of MCC values across datasets and splitting strategies. The largest reductions were observed for Tox21 (Table B.2). Random splitting achieved an average MCC of 0.305, whereas both full DataSAIL and scaled stratified DataSAIL yielded values of approximately 0.105.

The pronounced decrease in predictive performance coincided with the substantial reductions in leakage observed previously, suggesting that the optimistic performance estimates obtained under random splitting arise from similarity-induced information sharing between training and test partitions. Thus, a clear trade-off between leakage reduction and downstream predictive performance can be seen. Splitting strategies that more aggressively separated structurally similar molecules generally yielded lower MCC values, thereby producing more challenging evaluation settings. The reduced MCC in leakage-aware splitting shouldn't be interpreted as deterioration of model quality. It reflects a realistic out-of-distribution evaluation scenario.

Importantly, our scalable approach occupied an intermediate position between random and full DataSAIL. It preserved leakage-aware behaviour of DataSAIL splitting. This pattern is particularly evident for BBBP and HIV, suggesting that representative subset optimisation provides a practical compromise between rigorous evaluation and maintaining useful predictive signal.

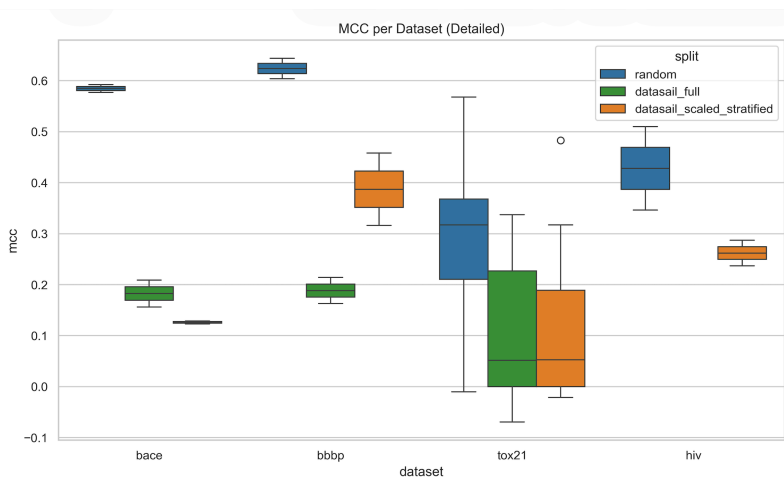


Figure 4- MCC per Dataset

Secondary Performance Metrics

The trends observed for MCC are largely reflected across the remaining evaluation metrics, including Accuracy, F1 score, ROC-AUC, Precision, Recall, and PR-AUC. These can be observed in Tables B.1-B.4. Random splitting generally yielded the most favourable predictive performance estimates, whereas full DataSAIL frequently produced lower scores across datasets.

The effects of leakage-aware splitting are especially pronounced for Tox21, where performance estimates decrease consistently across multiple metrics relative to random splitting. Given the strong reduction in leakage fraction observed for this dataset, these findings support the interpretation that conventional random partitions overestimate generalization performance in highly imbalanced molecular datasets.

6.4 Runtime

Runtime analysis compared the execution times of full DataSAIL, the random splitting approach and our proposed scalable approximation across the four datasets. The results are summarised in Table 4.

For the smaller datasets, BACE and BBBP, an immediate runtime advantage was not observed for the scalable approach. Full DataSAIL completed in 3.07 seconds for BACE and 3.75 seconds for BBBP, whereas the scaled approach required 4.81 and 5.78 seconds, respectively. Speedup factors of 0.64 and 0.65 in Table 4 indicate that the scalable approach was slower than the original implementation in these settings. This behaviour is expected and is attributed to the additional overhead introduced by representative subset construction and balanced k-nearest neighbour assignment. When the datasets are relatively small, the computational cost of full DataSAIL remains modest, and the overhead associated with the approximation framework outweighs the savings obtained from restricting optimisation to a subset of molecules.

A different pattern can be immediately noticed for Tox21. Full DataSAIL required 31.51 seconds, whereas our approach completed in 21.94 seconds, corresponding to a speedup factor of 1.44. Here, the reduction in optimization complexity began to compensate for the additional assignment overhead, resulting in a measurable improvement in computational efficiency.

The benefits become increasingly evident for HIV. Due to the substantially larger dataset size, full DataSAIL optimisation could not be completed within practical computational limits. However, the approximate runtime for the approach is 1800 seconds. The scaled DataSAIL successfully generated leakage-aware partitions in 297.98 seconds. This demonstrates that the representative subset approach substantially extends the applicability of similarity-aware splitting to datasets that would otherwise be computationally challenging to process using DataSAIL.

dataset	full_runtime_seconds	scaled_runtime_seconds	speedup
bace	3.07	5.75	0.53
bbbp	3.75	6.25	0.6
tox21	31.51	21.86	1.44
hiv		311.64	

Table 4- Runtime

7 DISCUSSION

The scaled DataSAIL preserves much of the leakage-aware behaviour of full DataSAIL while extending its applicability to larger datasets. The effectiveness of the approximation depends on dataset characteristics. Stratified downsampling improves class preservation and leakage reduction for the imbalanced Tox21 dataset but did not consistently outperform random downsampling across all benchmarks. These findings suggest that no single subset selection strategy is universally optimal.

The assignment of remaining samples introduced an additional trade-off between structural fidelity and split balance. The naïve assignment strategy, which is purely similarity-driven, naturally favoured dominant partitions, particularly the training set. This behaviour distorted validation and test distributions, ultimately reducing the reliability of downstream evaluation. The balanced kNN assignment, which used a balance-aware weighting mechanism and similarity score was therefore used. The approach dynamically corrects assignment bias while still preserving local structural consistency. Assignment quality depends on several settings, particularly the chosen neighbour parameter k , similarity thresholding behaviour, and balance correction formulation. This begs for dynamic hyperparameter tuning as different datasets may require different hyperparameter settings for optimal performance.

The evaluation step highlighted the challenging relationship between leakage reduction and predictive performance. Both approaches, DataSAIL and scaled DataSAIL, encouraged stronger structural separation, which consequently, produced more challenging evaluation settings, which lowered conventional performance metrics compared to random splitting^{9,10}. However, such reductions do not necessarily mean poorer performance. Instead, they reflect more realistic estimates of generalization to unseen chemical space.

Scalable runtime is one of the central motivations of our approach. Runtime analysis demonstrated that the scalable framework is not intended to replace DataSAIL for all datasets. While the additional assignment overhead made the approximation slower on smaller datasets, computational benefits emerged as dataset size increased, ultimately enabling leakage-aware splitting for HIV where full optimisation was impractical.

Future work will explore the evaluation of the approach on larger datasets to observe credible patterns. Subsequently, dynamic hyperparameter tuning for chosen neighbour parameter k would be explored. Additionally, extending the framework to directed message

passing neural networks and larger molecular benchmarks also represents an important direction for our future evaluation.

8 CONCLUSION

We explored a scalable extension of DataSAIL using representative subset optimization and balanced kNN assignment. The computational complexity of full similarity-based optimization leaves the space open for an approach that combines the benefits while being computationally accessible. Our scaled DataSAIL approach fits in by reducing the optimization space by applying DataSAIL exclusively to representative molecular subsets and using a balanced k-nearest neighbour assignment strategy to assign the remaining samples.

Our methodology, primarily, focused on preserving structural consistency, reducing inter-split leakage, and respecting target split distributions simultaneously. By combining similarity-weighted split voting with dynamic balance-aware correction factors, the dominance effects of naïve nearest neighbour assignment approach were mitigated.

Next, we explored subset selection strategies, including random and stratified sampling, and evaluated their impact on leakage reduction, split quality preservation, downstream machine learning performance, and runtime scalability across multiple molecular datasets. The effectiveness of the sampling strategy is dataset dependent, with stratified sampling proving particularly beneficial for the imbalanced Tox21 dataset but not consistently outperforming random downsampling across all datasets.

The preliminary results demonstrated that scaled DataSAIL retains much of the leakage-aware behaviour of full DataSAIL while substantially reducing information leakage relative to random splitting. Runtime analysis shows that scaled DataSAIL is not intended to replace full DataSAIL for all applications. The additional assignment overhead limited efficiency gains for smaller datasets, but the scaled DataSAIL becomes increasingly advantageous as dataset size increases and enables leakage-aware splitting for HIV, where full optimization was computationally impractical.

The experimental analysis is ongoing; the early research demonstrates a promising direction toward scalable leakage-aware splitting for large molecular datasets. As machine learning models continue to improve in predictive capacity, ensuring realistic and leakage-resistant evaluation protocols becomes increasingly critical for reliable scientific conclusions and practical deployment in drug discovery applications.

Future extensions on larger datasets are set to give credible insights and produce comparative analysis, which will help us in establishing the roadmap for exploration.

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APPENDIX A- ADDITIONAL SPLIT QUALITY RESULTS

This appendix presents supplementary split-quality evaluation results referenced in Section 6.2. These results provide additional detail regarding statistical representativeness and structural preservation of the generated dataset partitions. Distribution Gap (DG), Distribution Ratio Error (DRE), and Kullback–Leibler (KL) divergence quantify preservation of class distributions, while pairwise similarity preservation assesses the extent to which representative subsets retain the structural characteristics of the original molecular datasets.

dataset	split	mean	max
bace	datasail_scaled	0.132	0.21
bace	datasail_scaled_stratified	0.132	0.224
bbbp	datasail_scaled	0.045	0.08
bbbp	datasail_scaled_stratified	0.067	0.096
hiv	datasail_scaled	0.017	0.032
hiv	datasail_scaled_stratified	0.016	0.031
tox21	datasail_scaled	0.043	0.087
tox21	datasail_scaled_stratified	0.018	0.025

Table A.1- Distribution Gap

dataset	method	MeanDRE	MaxDRE
bace	datasail_scaled_random	0.289	0.459
bace	datasail_scaled_stratified	0.289	0.49
bbbp	datasail_scaled_random	0.059	0.105
bbbp	datasail_scaled_stratified	0.088	0.126
hiv	datasail_scaled_random	0.491	0.904
hiv	datasail_scaled_stratified	0.443	0.87
tox21	datasail_scaled_random	1.024	2.041
tox21	datasail_scaled_stratified	0.428	0.585

Table A.2- Distribution Ratio Error

dataset	method	MeanKL	MaxKL
bace	datasail_scaled_random	0.054216	0.103312
bace	datasail_scaled_stratified	0.057547	0.119858
bbbp	datasail_scaled_random	0.009535	0.02192
bbbp	datasail_scaled_stratified	0.017626	0.033222
hiv	datasail_scaled_random	0.006122	0.009665
hiv	datasail_scaled_stratified	0.003646	0.009057
tox21	datasail_scaled_random	0.020662	0.043574
tox21	datasail_scaled_stratified	0.006906	0.012841

Table A.3- KL Divergence

dataset	method	full_similarity	subset_similarity	delta_similarity
bace	random	0.165	0.166	0.001
bace	stratified	0.165	0.163	0.003
bbbp	random	0.104	0.104	0.0
bbbp	stratified	0.104	0.103	0.0
tox21	random	0.082	0.083	0.0
tox21	stratified	0.082	0.08	0.002
hiv	random	0.095	0.095	0.0
hiv	stratified	0.095	0.094	0.0

Table A.4- Pairwise Similarity Preservation

APPENDIX B- COMPLETE MACHINE LEARNING EVALUATION RESULTS

This appendix contains the complete machine learning evaluation results referenced in Section 6.3. While Matthews Correlation Coefficient (MCC) was used as the primary metric for interpretation in the main text, the tables presented here include the full set of performance measures, including Accuracy, F1 Score, ROC-AUC, Precision, Recall, and PR-AUC. These results provide a more comprehensive overview of model behaviour under the different dataset splitting strategies.

dataset	split	label	model	accuracy	f1	mcc	auc
bace	random	Class	random_forest	0.789	0.771	0.577	0.866
bace	random	Class	logistic_regression	0.796	0.78	0.592	0.826
bace	datasail_full	Class	random_forest	0.68	0.226	0.156	0.715
bace	datasail_full	Class	logistic_regression	0.687	0.356	0.209	0.646
bace	datasail_scaled_stratified	Class	random_forest	0.767	0.136	0.123	0.805
bace	datasail_scaled_stratified	Class	logistic_regression	0.65	0.36	0.129	0.615
bbbp	random	p_np	random_forest	0.868	0.917	0.644	0.92
bbbp	random	p_np	logistic_regression	0.854	0.904	0.604	0.878
bbbp	datasail_full	p_np	random_forest	0.847	0.917	0.163	0.719
bbbp	datasail_full	p_np	logistic_regression	0.842	0.912	0.214	0.755
bbbp	datasail_scaled_stratified	p_np	random_forest	0.876	0.932	0.316	0.767
bbbp	datasail_scaled_stratified	p_np	logistic_regression	0.876	0.929	0.458	0.757

Table B.1- ML evaluation BACE, BBBP

dataset	model	split	mcc_mean	mcc_std	auc_mean	auc_std	f1_mean	f1_std	accuracy_mean	accuracy_std
tox21	logistic_regression	datasail_full	0.14	0.14	0.63	0.12	0.18	0.16	0.91	0.07
tox21	logistic_regression	datasail_scaled_random	0.12	0.09	0.67	0.1	0.15	0.12	0.92	0.08
tox21	logistic_regression	datasail_scaled_stratified	0.12	0.12	0.61	0.11	0.16	0.12	0.92	0.06
tox21	logistic_regression	random	0.28	0.08	0.72	0.06	0.34	0.08	0.9	0.07
tox21	random_forest	datasail_full	0.07	0.11	0.7	0.13	0.05	0.09	0.94	0.06
tox21	random_forest	datasail_scaled_random	0.14	0.15	0.73	0.12	0.11	0.13	0.95	0.04
tox21	random_forest	datasail_scaled_stratified	0.09	0.16	0.72	0.11	0.06	0.13	0.96	0.03
tox21	random_forest	random	0.33	0.18	0.81	0.06	0.29	0.18	0.94	0.04

Table B.2- ML evaluation Tox21

dataset	split	label	model	accuracy	f1	mcc	auc	precision	recall	pr_auc
bace	random	Class	random_forest	0.789	0.771	0.577	0.866	0.75	0.794	0.834
bace	random	Class	logistic_regression	0.796	0.78	0.592	0.826	0.753	0.809	0.776
bace	datasail_full	Class	random_forest	0.68	0.226	0.156	0.715	0.583	0.14	0.551
bace	datasail_full	Class	logistic_regression	0.687	0.356	0.209	0.646	0.565	0.26	0.49
bace	datasail_scaled_stratified	Class	random_forest	0.767	0.136	0.123	0.805	0.5	0.079	0.485
bace	datasail_scaled_stratified	Class	logistic_regression	0.65	0.36	0.129	0.615	0.314	0.421	0.344
bbbp	random	p_np	random_forest	0.868	0.917	0.644	0.92	0.856	0.987	0.959
bbbp	random	p_np	logistic_regression	0.854	0.904	0.604	0.878	0.876	0.934	0.927
bbbp	datasail_full	p_np	random_forest	0.847	0.917	0.163	0.719	0.847	1.0	0.935
bbbp	datasail_full	p_np	logistic_regression	0.842	0.912	0.214	0.755	0.86	0.971	0.934
bbbp	datasail_scaled_stratified	p_np	random_forest	0.876	0.932	0.316	0.767	0.882	0.989	0.947
bbbp	datasail_scaled_stratified	p_np	logistic_regression	0.876	0.929	0.458	0.757	0.921	0.937	0.928

Table B.3- ML extended evaluation BACE, BBBP

tox21	logistic_regression	datasail_full	0.201	0.168	0.183	0.186	0.176	0.157
tox21	logistic_regression	datasail_scaled_random	0.185	0.148	0.169	0.152	0.136	0.076
tox21	logistic_regression	datasail_scaled_stratified	0.154	0.117	0.183	0.15	0.124	0.079
tox21	logistic_regression	random	0.348	0.107	0.335	0.074	0.284	0.102
tox21	random_forest	datasail_full	0.254	0.329	0.031	0.054	0.213	0.212
tox21	random_forest	datasail_scaled_random	0.393	0.412	0.067	0.084	0.205	0.146
tox21	random_forest	datasail_scaled_stratified	0.31	0.46	0.039	0.082	0.229	0.16
tox21	random_forest	random	0.692	0.324	0.189	0.125	0.406	0.137

Table B.4- ML extended evaluation Tox21

APPENDIX C- SUPPLEMENTARY FIGURES

This appendix contains supplementary figures that support the methodology described in the main text. Figure C.1 presents the simplified original DataSAIL workflow, The figure is provided for reference and contextual comparison with the proposed scalable approximation framework.



Figure C.1- DataSAIL Workflow

Figure C.2 illustrates the limitation of naïve nearest-neighbour assignment observed during preliminary experiments. Because the training partition contains the largest number of molecules, unseen samples frequently identify training molecules as their nearest neighbours. Repeated similarity-based assignment therefore causes the training partition to grow disproportionately, resulting in deviations from the intended train-validation-test proportions and reducing the quality of the final dataset split.

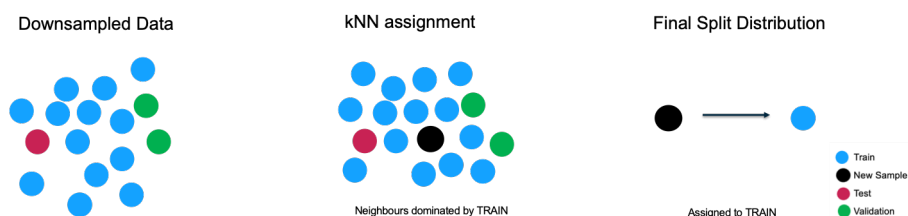


FIGURE C.2- Schema showing all dominant neighbours are Train which leads to repeated assignment which leads to growing imbalance

Figure C.3 presents the proposed balanced k-nearest neighbour assignment strategy. In addition to molecular similarity, assignment decisions incorporate balance-aware correction factors that favour underrepresented partitions and penalise oversized ones. This mechanism preserves structural consistency while maintaining the desired split ratios throughout the propagation process.

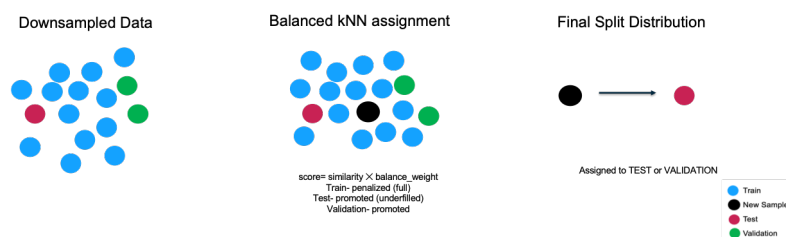


FIGURE C.3- Schema showing Balanced kNN assignment approach